

BPC_02_VRScreenFade

Component

1. Component Overview

Purpose

BPC_02_VRScreenFade is a VR screen fade component attached to the VR character Blueprint. It creates a full-screen plane in front of the camera and controls its opacity through the dynamic material's PlayerFade parameter, covering three fade scenarios: an automatic fade out when the game starts, a dynamic vignette while moving (anti-motion-sickness), and fades during teleportation or scene changes.

Naming conventions

- Fade Out = fade from black: the screen goes from black to clear (PlayerFade decreases, 1 to 0)
- Fade In = fade to black: the screen goes from clear to black (PlayerFade increases, 0 to 1)
- PlayerFade parameter: 0 = fully transparent, 1 = fully opaque (black)

Usage scenarios:

- Automatic fade out at game start, giving the player a smooth transition into the game
- Dynamic movement vignette (anti-motion-sickness): while the player moves, the visible screen area shrinks based on movement speed to reduce VR motion sickness

- Fades during teleportation/scene changes

2. Core Variables

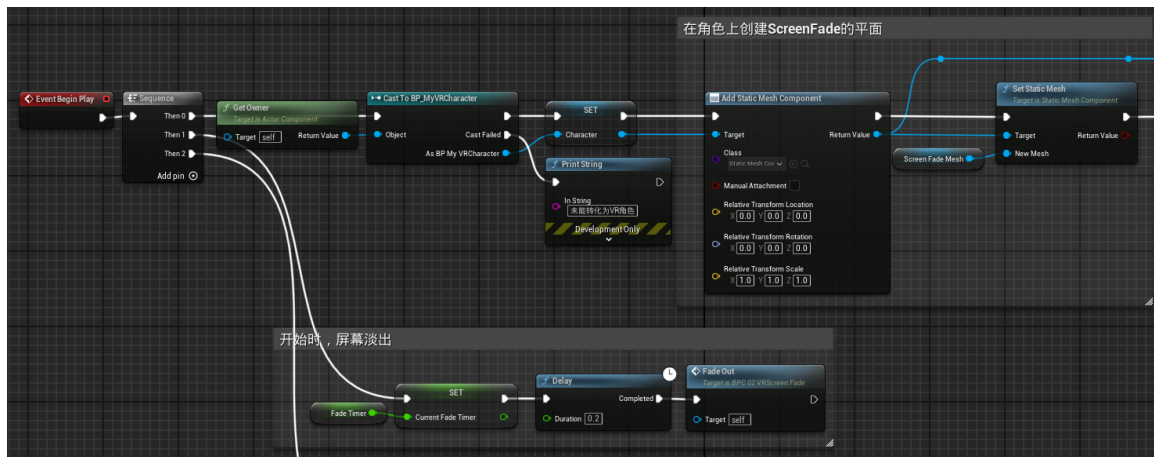
Variable	Type	Description
ScreenFadeMesh	Static Mesh component reference	Mask plane mesh asset
ScreenFadeTrans	Transform	Plane transform relative to the camera
Color	Linear Color	Mask color
FadeTimer	Float	Total fade duration (seconds)

3. Initialization (BeginPlay)

Description

During initialization the component does three things: obtain the VR character reference, create the mask plane and attach it to the camera, and create the dynamic material instance and set the initial color. Afterwards it automatically runs Fade Out after a 0.2-second delay (the screen goes from black to clear at game start).

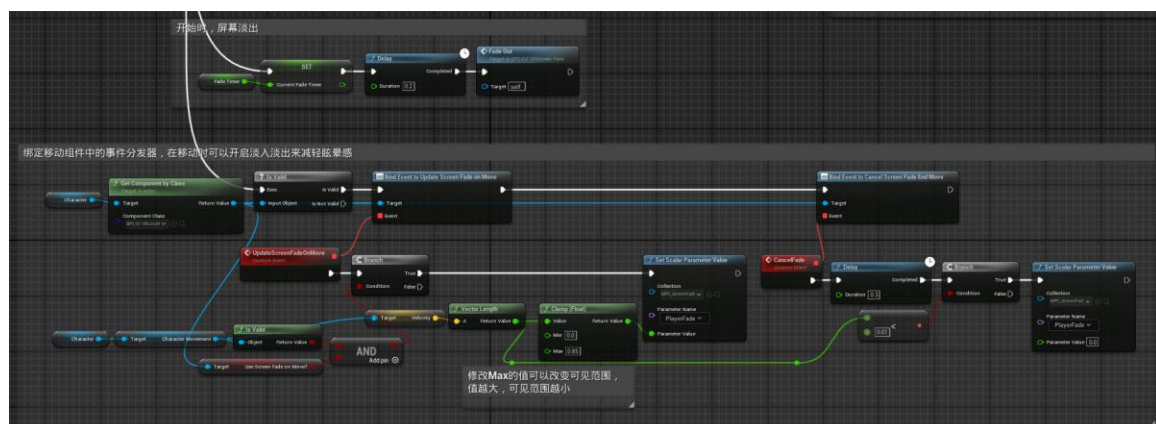
Blueprint screenshot - getting the character reference and creating the plane



- The 0.2-second delay waits for the level to finish loading, avoiding screen flicker during loading

Description

Blueprint screenshots



Flow

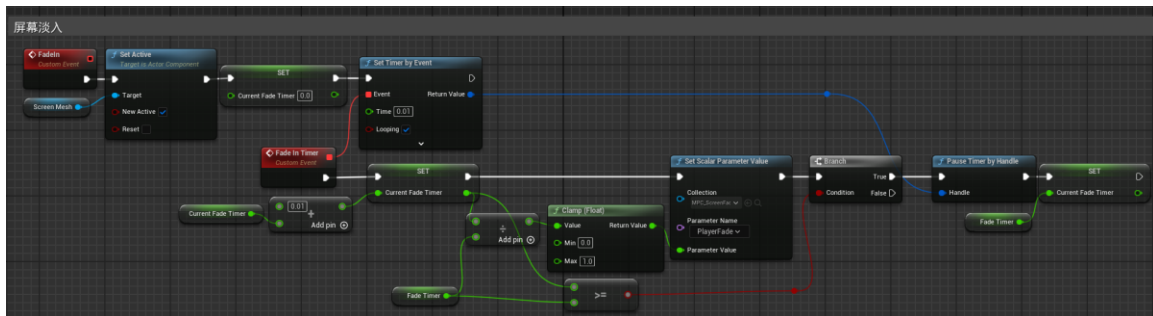
- When finished, Set Active (false) hides the plane to reduce rendering cost

6. Fade In Function (fade to black, the screen becomes black)

Description

The gradient effect of the screen going from clear to black, usually called when a teleport starts or before a scene change.

Blueprint screenshots



Key points

- Increasing direction: Current Fade Time counts up from 0 to FadeTime, and PlayerFade goes from low to high (the screen goes from clear to black)
- Set Active (true) at the start ensures the plane is visible